SCRUM FRAMEWORK

COMP6100001 - SOFTWARE ENGINEERING
SESSION 04

UNIVERSITAS **BINA NUSANTARA**

SUBJECT MATTER EXPERT

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SESSION 04 – SCRUM FRAMEWORK

LEARNING OBJECTIVES

At the end of this session, students will be expected to be able to:

- Explain the Core Components of the Scrum Framework
- Analyze the Benefits and Challenges of Scrum Implementation

SESSION 04 – SCRUM FRAMEWORK

CONTENT OUTLINE

- Scrum framework overview
- Scrum roles
- Scrum artifacts
- Scrum events
- User stories
- Benefits & challenges of Scrum

AGILE FRAMEWORK: SCRUM

SCRUM

- Scrum is a lightweight framework that helps people, teams and organizations generate value through adaptive solutions for complex problems
- The Scrum framework is purposefully incomplete, only defining the parts required to implement Scrum theory
- Scrum is built upon by the collective intelligence of the people using it. Rather than provide people with detailed instructions, the rules of Scrum guide their relationships and interactions.
- Various processes, techniques and methods can be employed within the framework. Scrum wraps around existing practices or renders them unnecessary. Scrum makes visible the relative efficacy of current management, environment, and work techniques, so that improvements can be made.

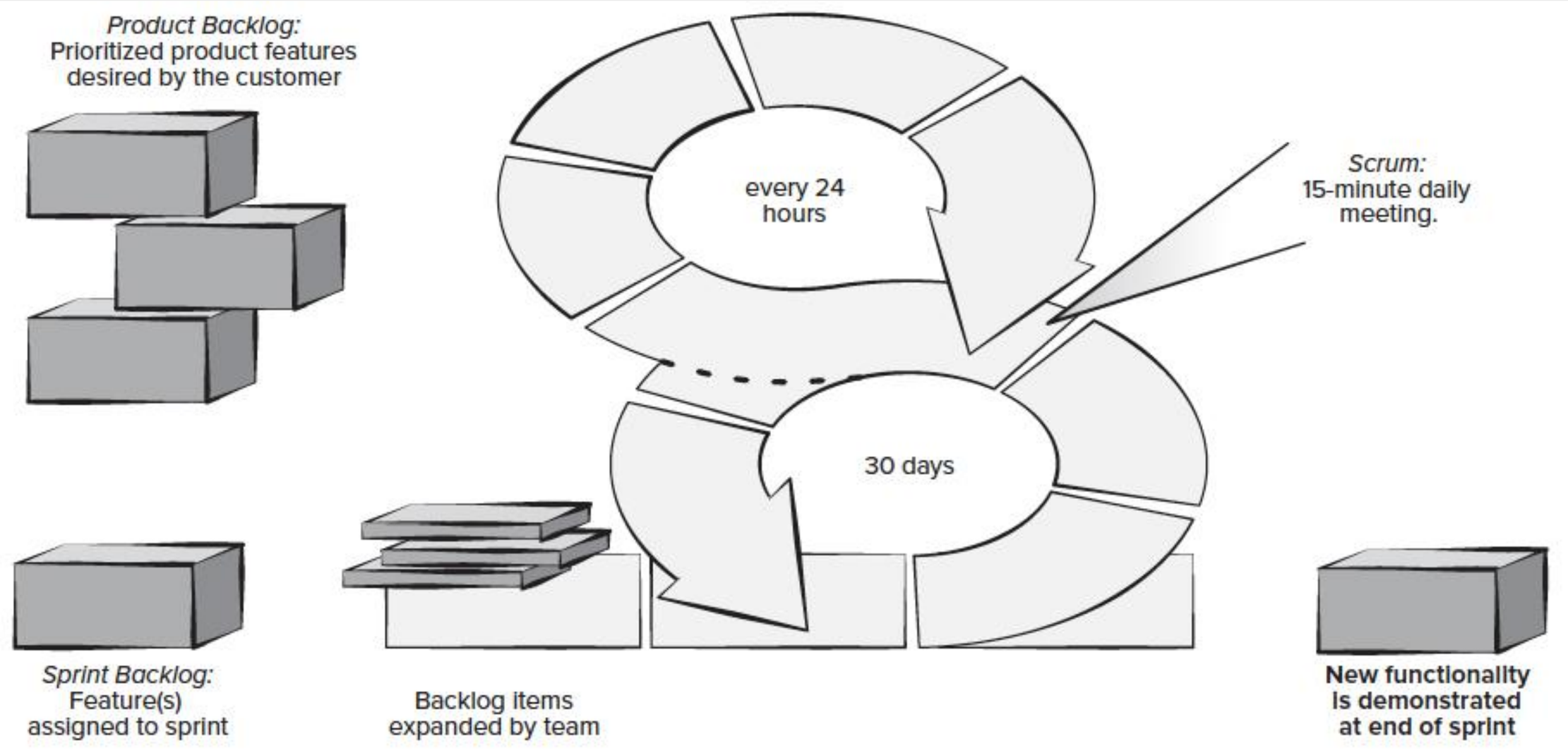
SCRUM

FUNDAMENTAL AND PILLARS

- Scrum is founded on **empiricism** and **lean thinking**
- **Empiricism** → asserts that knowledge comes from experience and making decisions based on what is observed
- **Lean thinking** → reduces waste and focuses on the essentials
- Scrum employs an iterative, incremental approach to optimize predictability and to control risk
- Empirical Scrum pillars → **transparency**, **inspection**, and **adaptation**

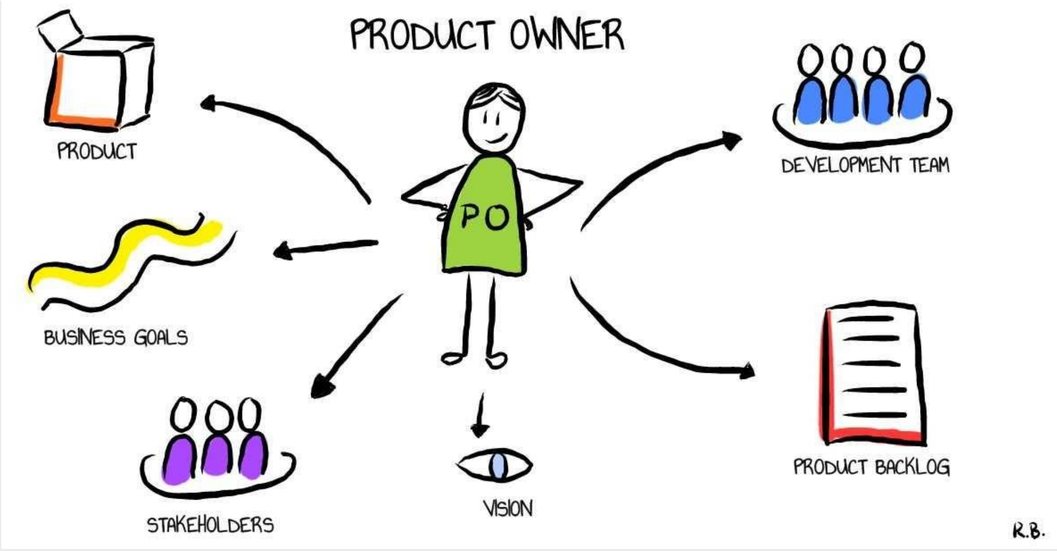
AGILE FRAMEWORK: SCRUM

SCRUM



SCRUM ROLES

PRODUCT OWNER



Responsibilities:

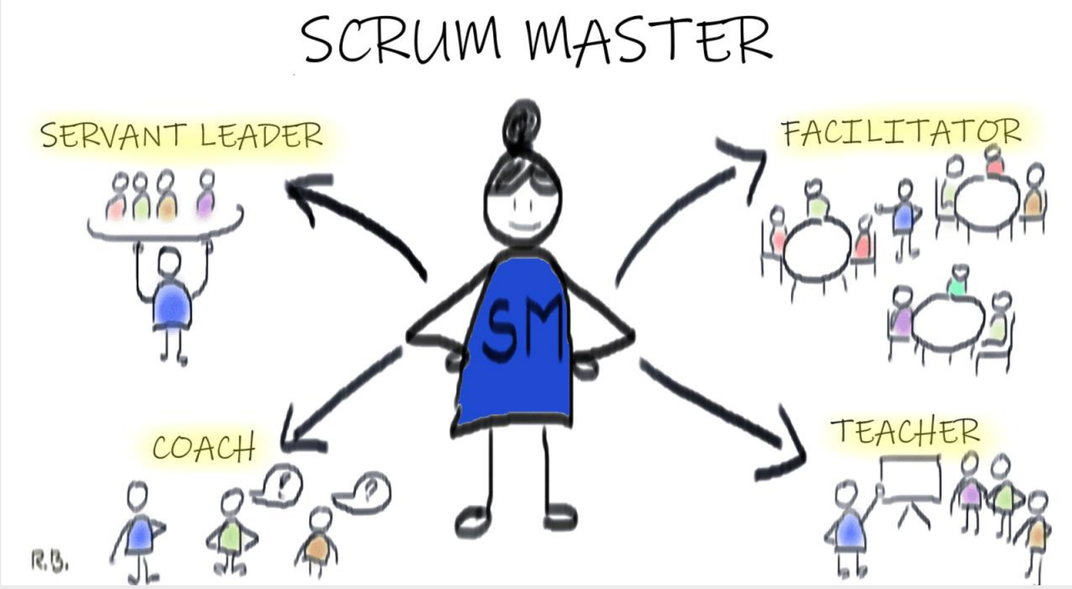
- Defines the product vision.
- Manages the Product Backlog.
- Prioritizes needs based on stakeholder input and business value.
- Ensures the team delivers value to the business.

Skills Needed:

- Strong communication and negotiation skills.
- Understanding of customer needs and market trends.

SCRUM ROLES

SCRUM MASTER



Responsibilities:

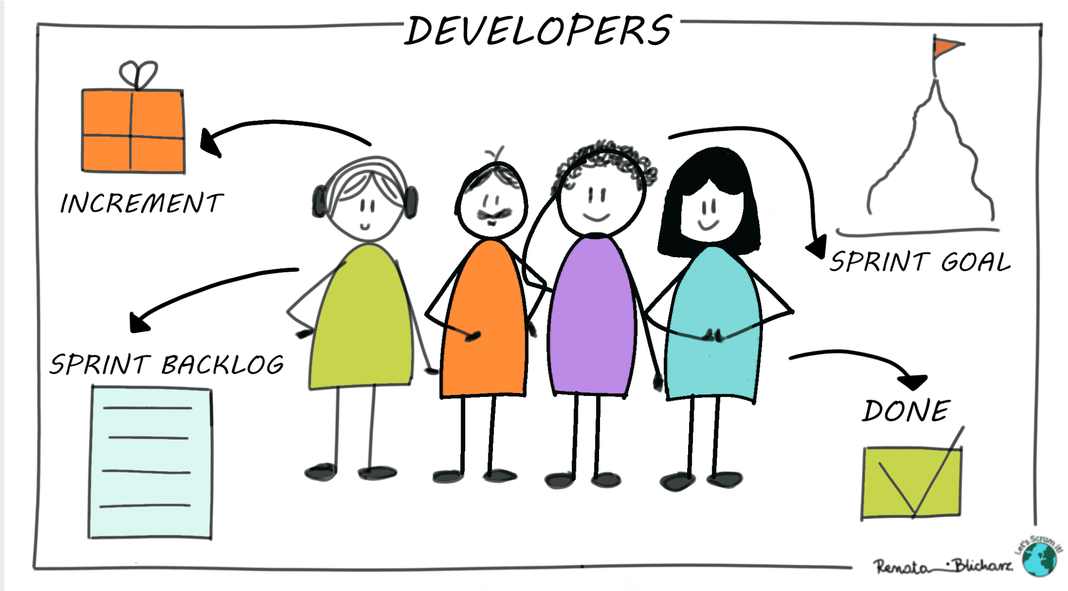
- Facilitates Scrum events and processes.
- Removes impediments to the team's progress.
- Coaches the team in Scrum practices.
- Serves as a servant leader to the Development Team.

Skills Needed:

- Knowledge of Agile and Scrum principles.
- Strong facilitation and conflict resolution skills.

SCRUM ROLES

DEVELOPMENT TEAM



Responsibilities:

- Self-organizing professionals who do the work of delivering a potentially releasable Increment.
- Cross-functional with all skills necessary to create the product Increment.

Characteristics:

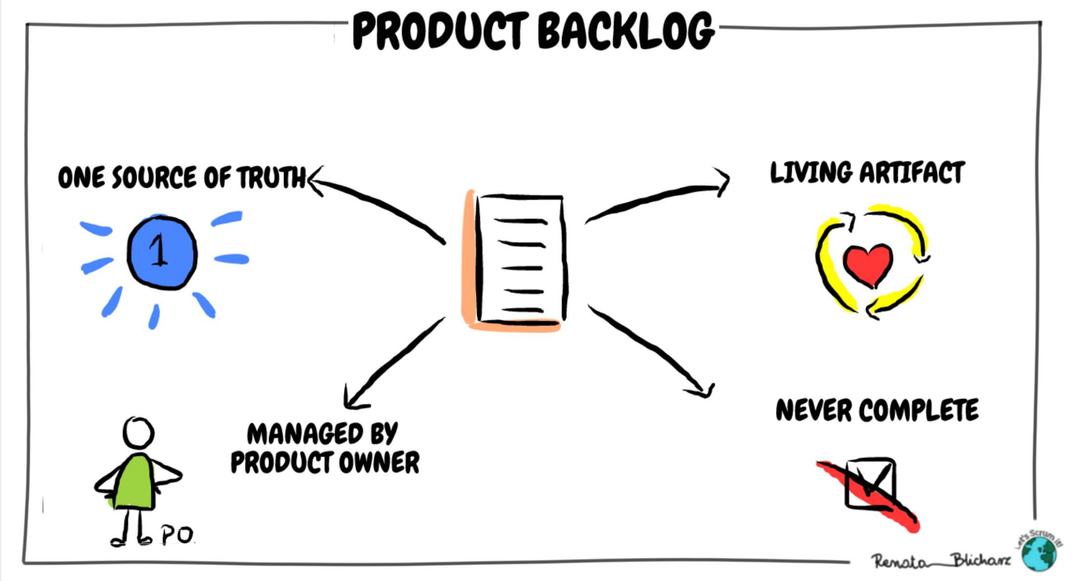
- Typically consists of 3-9 members.
- Collective ownership and accountability.

SCRUM ARTIFACTS

PRODUCT BACKLOG

- The **product backlog** is all the features for the product, an ordered list of everything that is known to be needed in the product.

- **Characteristics**:
  - Dynamic; evolves as the product and environment do.
  - Items include features, functions, requirements, enhancements, and fixes.

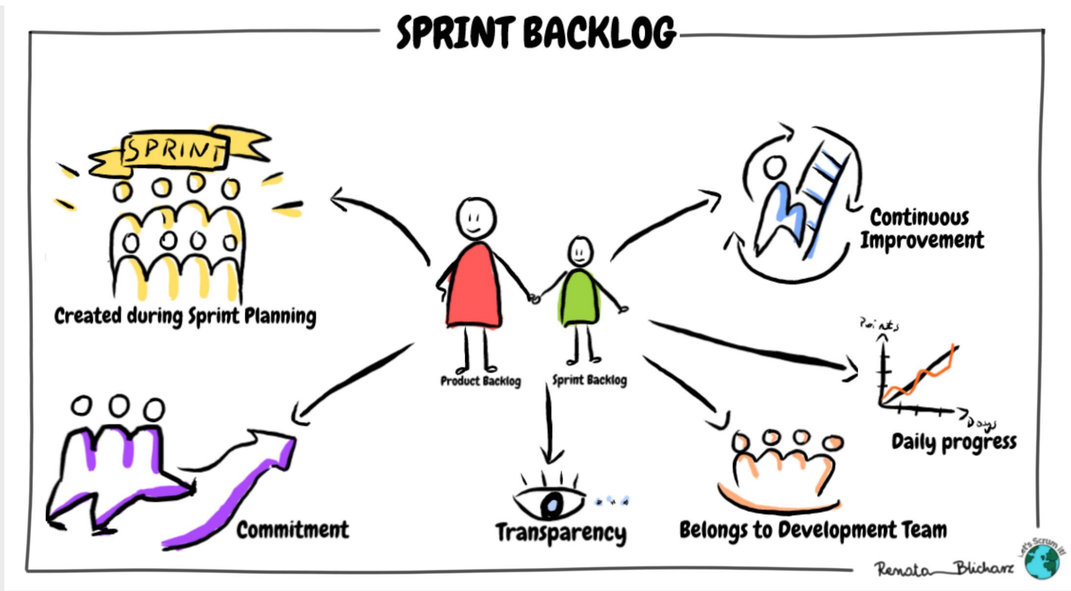


SCRUM ARTIFACTS

SPRINT BACKLOG

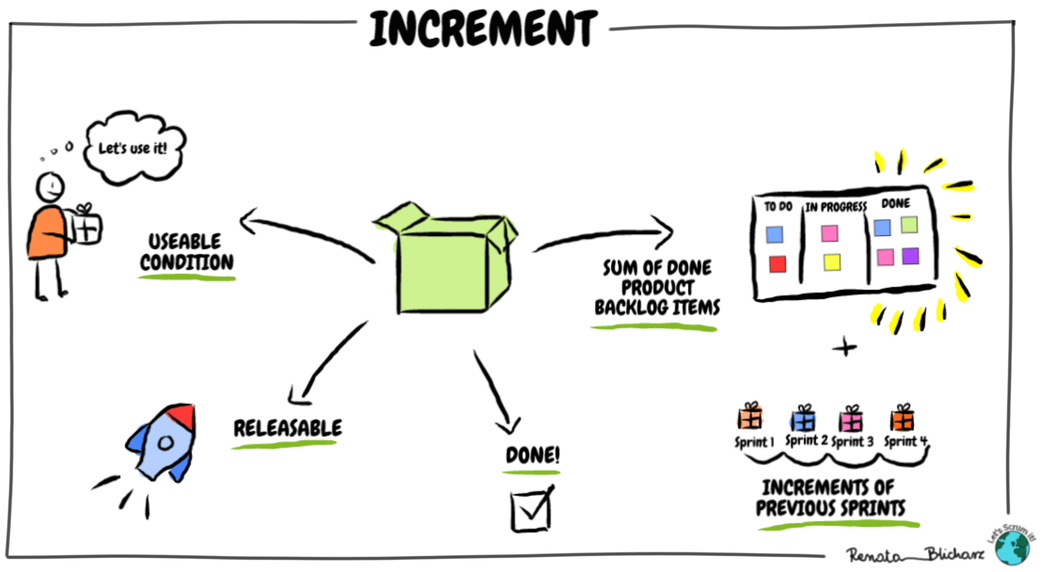
- The **sprint backlog** is all the features that will be worked on for that sprint, the set of Product Backlog items selected for the Sprint, plus a plan for delivering the product Increment.

- **Characteristics**:
  - Created during Sprint Planning.
  - Owned and managed by the Development Team.
- These should be broken down into discrete tasks:
  - Fine-grained
  - Estimated
  - Assigned to individual team members
  - Acceptance criteria should be defined
- User stories are often used



SCRUM ARTIFACTS

INCREMENT



- The sum of all the Product Backlog items completed during a Sprint and the value of the increments of all previous Sprints.

- **Characteristics**:
  - Must be in useable condition regardless of whether the Product Owner decides to release it.

SCRUM EVENTS

SCRUM MEETINGS

- Sprint Planning Meeting
  - Entire Team decides together what to tackle for that sprint
- Daily Scrum Meeting
  - Quick Meeting to touch base on:
    - What have I done? What am I doing next? What am I stuck on/need help?
- Sprint Retrospective
  - Review sprint process
- Sprint Review Meeting
  - Review product

SCRUM EVENTS

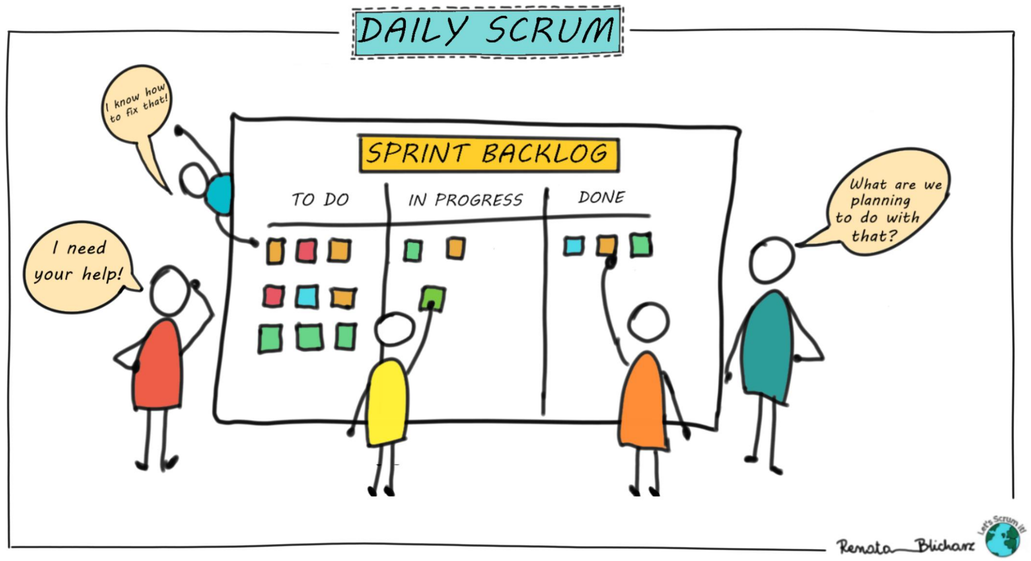
SPRINT PLANNING

- **Purpose**:
  - Define what can be delivered in the Sprint.
  - Plan how the work will be achieved.
- **Time-box**:
  - Maximum of 8 hours for a one-month Sprint.
- **Activities**:
  - **What**: The Product Owner presents the desired Product Backlog items.
  - **How**: The Development Team plans the work necessary to deliver the Increment.

SCRUM EVENTS

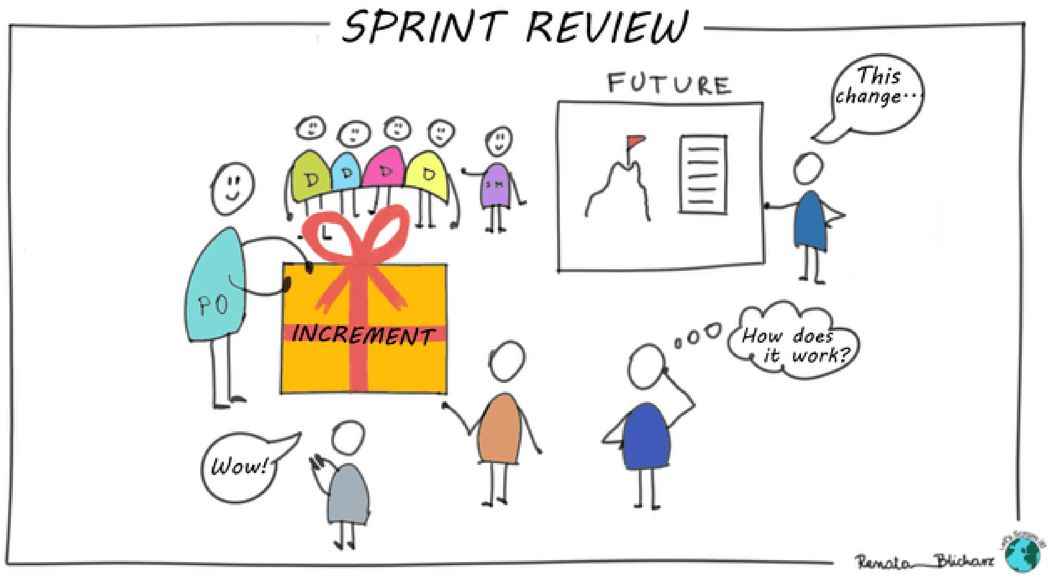
DAILY STAND-UP (DAILY SCRUM)

- **Purpose**:
  - Synchronize activities and create a plan for the next 24 hours.
- **Time-box**:
  - 15 minutes.
- **Questions Addressed**:
  - What did I do yesterday that helped the Development Team meet the Sprint Goal?
  - What will I do today to help the Development Team meet the Sprint Goal?
  - Do I see any impediment that prevents me or the Development Team from meeting the Sprint Goal?



SCRUM EVENTS

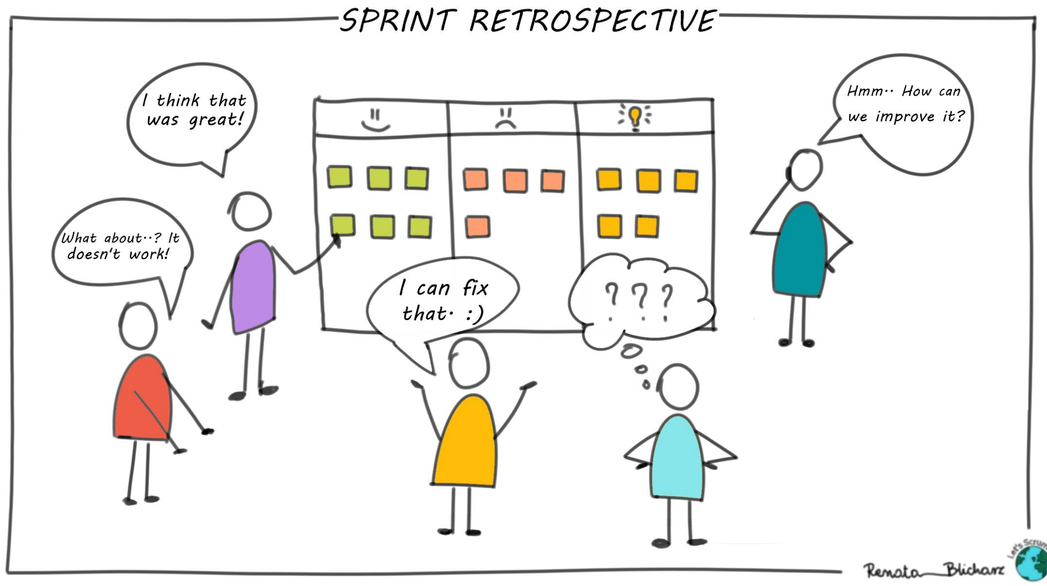
SPRINT REVIEW



- **Purpose**:
  - Inspect the Increment and adapt the Product Backlog if needed.
- **Time-box**:
  - Maximum of 4 hours for a one-month Sprint.
- **Activities**:
  - Demonstration of the Increment.
  - Feedback from stakeholders.
  - Discussion of what to do next.

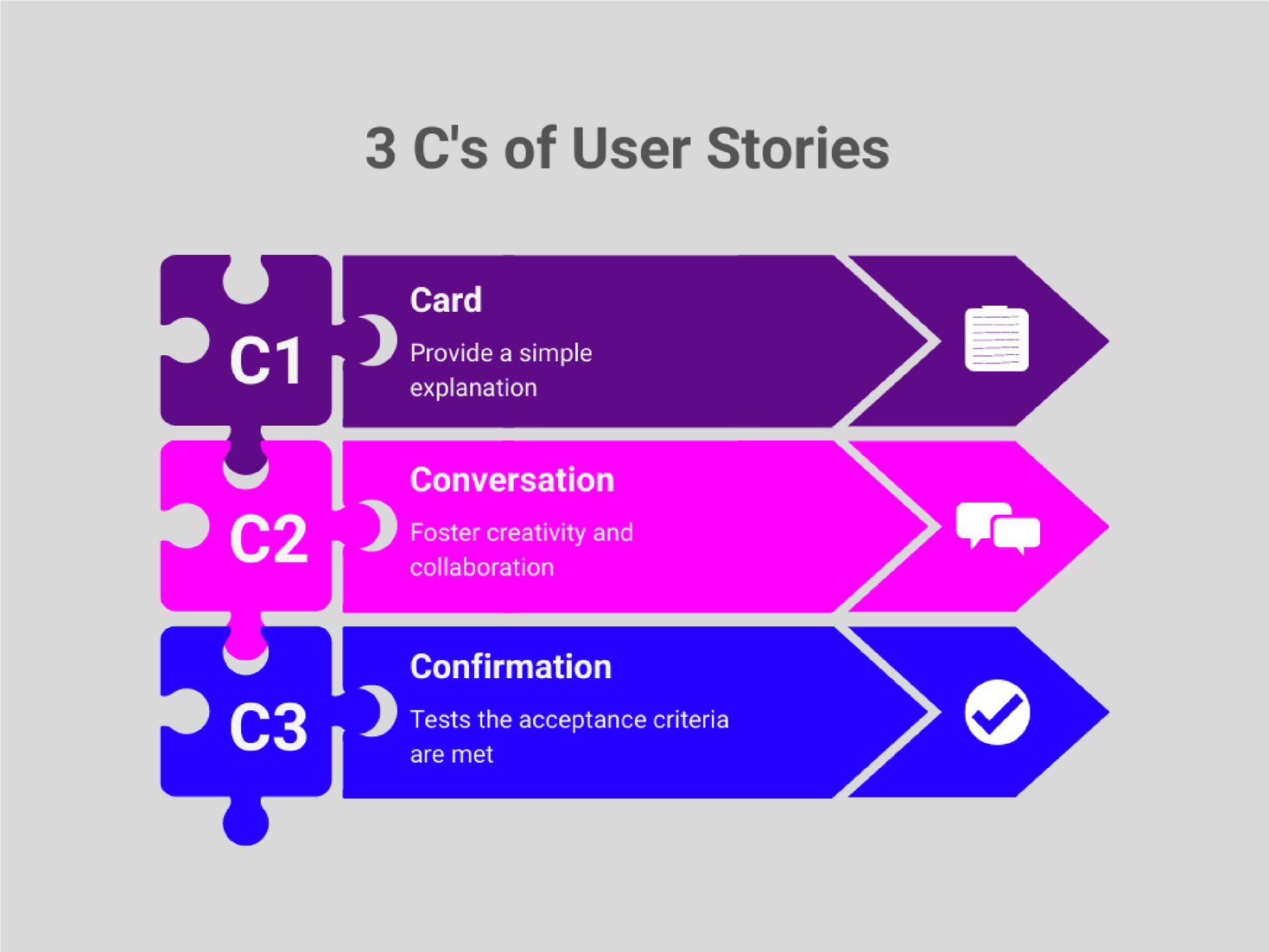
SCRUM EVENTS

SPRINT RETROSPECTIVE



- **Purpose**:
  - Plan ways to increase quality and effectiveness.
- **Time-box**:
  - Maximum of 3 hours for a one-month Sprint.
- **Activities**:
  - Inspect how the last Sprint went regarding people, relationships, processes, and tools.
  - Identify and order major items that went well and potential improvements.
  - Create a plan for implementing improvements.

USER STORIES



USER STORIES

CARD — A SHORT REMINDER OF THE REQUIREMENT

- The **Card** is a brief written description
- It captures **who, what, and why**
- It is **not** a detailed specification

"As a [role], I want [function], so that [value]"

Examples:

1. As a **student**, I want to **see my grades**, so that **I can track my academic progress.**
2. As a **customer**, I want to **reset my password**, so that **I can regain account access**.

USER STORIES

CONVERSATION — SHARED UNDERSTANDING THROUGH DISCUSSION

- Details emerge through **conversation**
- Developers, Product Owner, and stakeholders collaborate
- Questions clarify assumptions and edge cases

Examples:

1. For “see my grades”:
    - Should grades be real-time?
    - Who can view them?
    - Can grades be downloaded?
2. For “reset password”:
    - Email or SMS?
    - Expiration time?
    - Security verification needed?

USER STORIES

CONFIRMATION

- Defines **acceptance criteria**
- Ensures the story meets expectations
- Used to verify completion

Examples:

1. Grade viewing is done when:
    - Student sees grades for all enrolled courses
    - Only the student can access them
    - Data matches the official record
2. Password reset is done when:
    - Reset link expires after 10 minutes
    - User sets a new password successfully
    - Old password no longer works

USER STORIES

HOW TO EVALUATE USER STORY?

INVEST IN GOOD USER STORIES

Independent

Stories should be independent in a way that allows the easy prioritization and not overlap each other, even though it is not always possible.

Negotiable

Story is viewed as an invitation for conversation capturing what is desired, not a contract for features. It may acquire notes, test ideas etc. in order to meet a customer needs.

Valuable

Value of a story is based on customer needs, what is an important factor while prioritizing backlog. Value needs to be considered especially when splitting stories so each story brings a value to the customer.

Estimable

A good story can be estimated, meaning that developers can understand it and be able to implement it. Based on that it is possible to rank and schedule implementation.

Small

Stories tend to be small. Usually at most a few person-weeks of work or one iteration length. Smaller stories are easier to estimate and further details can be added later.

Testable

Common characteristic of good requirements is testability. This applies for stories as well. Testability is also an indication of understanding the goal of the story.

ACRONYM

BENEFITS AND CHALLENGES OF IMPLEMENTING SCRUM

BENEFITS AND CHALLENGES

Benefits:

- **Increased Transparency**: Regular meetings and artifacts make progress visible.
- **Flexibility and Adaptability**: Ability to respond to changes quickly.
- **Improved Quality**: Continuous testing and integration.
- **Higher Customer Satisfaction**: Regular feedback and involvement.
- **Team Empowerment**: Teams are self-organizing and make decisions.

Challenges:

- **Cultural Shift**: Requires a change in mindset and company culture.
- **Role Misunderstandings**: Confusion over responsibilities can hinder progress.
- **Resistance to Change**: Team members or management may resist new processes.
- **Requires Discipline**: Consistency in following Scrum practices is essential.
- **Scaling Issues**: Applying Scrum to large projects or organizations can be complex.

REFERENCES

- Roger Pressman and Bruce Maxim. (2020). ***Software Engineering: A Practitioner's Approach***. 9th E. McGraw- Hill. New York. ISBN: 9781259872976.